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Committee Member Two

Committee Member Three

Committee in Charge

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Your dedication here.

Acknowledgements

Your acknowledgments go here.

Abstract

Your abstract goes here. This should be a concise summary of the dissertation's main contributions, typically one to two paragraphs.

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Chapter 1

Introduction

This dissertation investigates [your topic]. The central question is [your question]. We approach this through [your methodology].

The framework of modal logic provides the formal backbone for this investigation. Kripke's seminal work [4] established the standard relational semantics, but the generalization to neighborhood semantics [6, 5] opens up a richer landscape of logics—including non-normal logics that lack monotonicity or other structural properties commonly assumed in the Kripke paradigm [2].

1.1 Motivations

Your motivations here. Explain why this topic matters and what gap in the literature the dissertation addresses.

1.2 Chapter Overview

In Chapter 2, we develop the formal preliminaries needed for the rest of the dissertation.

In Chapter 3, we present the main technical development.

In Chapter 4, we establish the central results.

Chapter 5 outlines directions for future work.

Chapter 2

First Chapter Title

This chapter lays the formal groundwork for the rest of the dissertation. We fix notation and recall the key definitions and results from modal logic that will be needed in later chapters.

2.1 Propositional Modal Logic

We work in a propositional modal language $\mathscr{L}^{\mathrm{P}}$ built from a countable set of propositional variables $\mathsf{Prop} = \{p, q, r, \ldots\}$, the Boolean connectives, and a unary modal operator $\Box$ (necessity). The dual $\Diamond$ (possibility) is defined as $\Diamond A \coloneqq \neg\Box\neg A$.

Definition 2.1.1. A *Kripke frame* is a pair $\mathfrak{F}_R = \langle W, R\rangle$ where W is a non-empty set of worlds and $R \subseteq W \times W$ is an accessibility relation.

Definition 2.1.2. A *Kripke model* $\mathfrak{M}_R = \langle W, R, V\rangle$ consists of a Kripke frame $\mathfrak{F}_R$ together with a valuation $V : \mathsf{Prop} \to \wp(W)$.

Truth at a world w in a model $\mathfrak{M}_R$ is defined recursively. The key clause for the modal operator is:

$$\mathfrak{M}_R, w \vDash \Box A \leftrightarrow \text{for all } v \in W, \text{ if } wRv \text{ then } \mathfrak{M}_R, v \vDash A.$$

We write $[\![A]\!]_R^{\mathfrak{M}} = \{w \in W : \mathfrak{M}_R, w \vDash A\}$ for the *truth set* of A in $\mathfrak{M}_R$.

2.2 Normal Modal Logics

The smallest normal modal logic, **K**, is axiomatized over classical propositional logic by:

K	$\Box(A \to B) \to (\Box A \to \Box B)$
Nec	From A, infer $\Box A$.

Extensions of **K** are obtained by adding axioms:

D	$\Box A \to \Diamond A$
U	$\Box A \to A$

These yield familiar logics: $\mathbf{D} = \mathbf{K} + \mathbf{D}$, $\mathbf{T} = \mathbf{K} + \mathbf{U}$, and so on through **S4** and **S5**.

THEOREM 2.2.1. **K** is sound and complete with respect to the class of all Kripke frames.

Proof. Soundness is routine. For completeness, construct the canonical model $\mathfrak{M}^{\mathcal{L}} = \langle W^{\mathcal{L}}, R^{\circ}, V^{\mathcal{L}} \rangle$ where $W^{\mathcal{L}}$ is the set of all maximal **K**-consistent sets, and $\Gamma R^{\circ} \Delta$ iff $\{A : \Box A \in \Gamma\} \subseteq \Delta$. The Lindenbaum–Tarski construction guarantees that every consistent formula is satisfiable in $\mathfrak{M}^{\mathcal{L}}$. ■

2.3 Neighborhood Semantics

Neighborhood semantics generalize relational semantics by replacing the accessibility relation with a neighborhood function [6, 5].

DEFINITION 2.3.1. A *neighborhood frame* is a pair $\mathfrak{F}_N = \langle W, f_N \rangle$ where W is a non-empty set and $f_N : W \to \wp(\wp(W))$ assigns to each world a collection of neighborhoods.

DEFINITION 2.3.2. A *neighborhood model* $\mathfrak{M}_N = \langle W, f_N, V \rangle$ consists of a neighborhood frame together with a valuation.

The truth clause for the modal operator becomes:

$$\mathfrak{M}_N, w \vDash \Box A \leftrightarrow [\![A]\!]^{\mathfrak{M}_N} \in f_N(w).$$

Remark 2.3.3. Neighborhood semantics can model *non-normal* logics that lack the **K** axiom or the necessitation rule. This makes them suitable for deontic and other weaker-than-normal modal logics. ◁

Chapter 3

Second Chapter Title

This chapter develops the main technical apparatus of the dissertation. Building on the neighborhood semantics introduced in Chapter 2, we specialize the framework to deontic logic and extend it to conditional operators.

3.1 Deontic Neighborhood Frames

We now interpret the modal operator as obligation. In place of the generic $\Box$, we write O for "it ought to be that" and P for "it is permissible that," where $\mathsf{P}A := \neg\mathsf{O}\neg A$.

Definition 3.1.1. A *deontic neighborhood frame* is a neighborhood frame $\mathfrak{F}_N = \langle W, f_N \rangle$ where $f_N(w)$ is interpreted as the collection of obligation-neighborhoods at w.

Frame conditions on f_N correspond to axioms of the logic:

Proposition 3.1.2. A neighborhood frame validates **D** (i.e., $\mathsf{O}A \to \mathsf{P}A$) if and only if f_N satisfies:

$$\text{for all } w \in W, \text{ if } X \in f_N(w) \text{ then } W \setminus X \notin f_N(w).$$

Proof. ($\Rightarrow$) Suppose the frame validates **D** and $X \in f_N(w)$. Let $V(p) = X$. Then $w \vDash \mathsf{O}p$, so by **D**, $w \vDash \mathsf{P}p$, i.e., $w \nvDash \mathsf{O}\neg p$. Hence $[\![\neg p]\!] = W \setminus X \notin f_N(w)$.

($\Leftarrow$) Assume the frame condition holds and $w \vDash \mathsf{O}A$. Then $[\![A]\!] \in f_N(w)$, so $W \setminus [\![A]\!] = [\![\neg A]\!] \notin f_N(w)$, giving $w \nvDash \mathsf{O}\neg A$, i.e., $w \vDash \mathsf{P}A$. ■

3.2 Monotonicity and Congruentiality

A key structural distinction is between *monotone* and merely *congruential* logics.

Definition 3.2.1. A neighborhood function f_N is *supplemented* (monotone) if for all $w \in W$:

$$X \in f_N(w) \text{ and } X \subseteq Y \implies Y \in f_N(w).$$

Note 3.2.2. Supplementation corresponds to the monotonicity rule (RM): from $\vdash A \to B$ infer $\vdash \mathsf{O}A \to \mathsf{O}B$. Logics that validate the congruence rule RE — from $\vdash A \leftrightarrow B$ infer $\vdash \mathsf{O}A \leftrightarrow \mathsf{O}B$ — without validating RM are called *congruential* or *classical*. ◁

Lemma 3.2.3. Every neighborhood function f_N has a unique smallest monotone extension ${f_N}^+$, defined by

$${f_N}^+(w) = \{Y \subseteq W : X \subseteq Y \text{ for some } X \in f_N(w)\}.$$

Proof. Clearly $f_N(w) \subseteq {f_N}^+(w)$. To see that ${f_N}^+$ is supplemented, suppose $Y \in {f_N}^+(w)$ and $Y \subseteq Z$. Then there exists $X \in f_N(w)$ with $X \subseteq Y \subseteq Z$, so $Z \in {f_N}^+(w)$. Minimality follows by construction. ■

3.3 Conditional Operators

For dyadic deontic logics, we move to a conditional obligation operator $\Rightarrow\!\mathsf{O}$, read "given A, it ought to be that B."

Definition 3.3.1. A *dyadic neighborhood frame* is a structure $\langle W, f_D, V\rangle$ where $f_D : W \times \wp(W) \to \wp(\wp(W))$ is a selection function mapping a world and a proposition to a collection of neighborhoods.

The truth clause for conditional obligation is:

$$\mathfrak{M}_N, w \vDash \mathrel{\Rightarrow\!\!\bigcirc}(A, B) \leftrightarrow [\![B]\!] \in f_D(w, [\![A]\!]).$$

THEOREM 3.3.2. A dyadic neighborhood frame validates $\mathrel{\Rightarrow\!\!\bigcirc}(A, B \wedge C) \to \mathrel{\Rightarrow\!\!\bigcirc}(A, B)$ if and only if f_D is monotone in the consequent.

Proof sketch. The proof follows the same pattern as Proposition 3.1.2: the left-to-right direction extracts the frame condition via a suitable valuation, and the right-to-left direction verifies the axiom directly. □

Chapter 4

Third Chapter Title

This chapter establishes the main completeness results for the logics developed in Chapters 2 and 3. The central technique is the construction of canonical neighborhood models.

4.1 Canonical Models

Convention 4.1.1. Throughout this chapter, L denotes a logic extending the minimal congruential (classical) modal logic $\mathbf{E}$, in the sense of Chellas [2]. $\triangleleft$

Definition 4.1.2. The *canonical neighborhood model* for L is $\mathfrak{M}^{\mathcal{L}} = \langle W^{\mathcal{L}}, f^{\mathcal{L}}, V^{\mathcal{L}} \rangle$ where:

(i) $W^{\mathcal{L}}$ is the set of all maximal L-consistent sets of formulas;

(ii) $V^{\mathcal{L}}(p) = \{\Gamma \in W^{\mathcal{L}} : p \in \Gamma\}$ for each $p \in \mathsf{Prop}$;

(iii) $f^{\mathcal{L}}(\Gamma) = \{[A] : \mathsf{O}A \in \Gamma\}$, where $[A] = \{\Delta \in W^{\mathcal{L}} : A \in \Delta\}$ is the *proof set* of A.

Observation 4.1.3. In the canonical model, truth sets and proof sets coincide: $[\![A]\!]^{\mathfrak{M}^{\mathcal{L}}} = [A]$ for every formula A. This is established by induction on formula complexity. $\triangleleft$

4.2 The Truth Lemma and Completeness

THEOREM 4.2.1 (Truth Lemma). For every formula A and every $\Gamma \in W^{\mathcal{L}}$:

$$\mathfrak{M}^{\mathcal{L}}, \Gamma \vDash A \leftrightarrow A \in \Gamma.$$

Proof. By induction on the complexity of A. The base case and Boolean cases are standard. For the modal case $A = \mathsf{O}B$:

($\Rightarrow$) If $\mathfrak{M}^{\mathcal{L}}, \Gamma \vDash \mathsf{O}B$, then $[\![B]\!]^{\mathfrak{M}^{\mathcal{L}}} \in f^{\mathcal{L}}(\Gamma)$. By Observation 4.1.3, $[B] \in f^{\mathcal{L}}(\Gamma)$, so by definition of $f^{\mathcal{L}}$, $\mathsf{O}B \in \Gamma$.

($\Leftarrow$) If $\mathsf{O}B \in \Gamma$, then $[B] \in f^{\mathcal{L}}(\Gamma)$. By the induction hypothesis, $[\![B]\!]^{\mathfrak{M}^{\mathcal{L}}} = [B]$, so $[\![B]\!]^{\mathfrak{M}^{\mathcal{L}}} \in f^{\mathcal{L}}(\Gamma)$, i.e., $\mathfrak{M}^{\mathcal{L}}, \Gamma \vDash \mathsf{O}B$. ■

COROLLARY 4.2.2. Every logic $L \supseteq \mathbf{E}$ is complete with respect to its canonical neighborhood model: if A is valid in all neighborhood models for L, then $A \in L$.

Proof. Suppose $A \notin L$. Then $\{\neg A\}$ is L-consistent, so by Lindenbaum's Lemma it extends to some $\Gamma \in W^{\mathcal{L}}$ with $\neg A \in \Gamma$. By the Truth Lemma, $\mathfrak{M}^{\mathcal{L}}, \Gamma \vDash \neg A$, so A is not valid in $\mathfrak{M}^{\mathcal{L}}$. ■

4.3 Monotonicity and Canonicity

Not every frame condition is preserved in canonical models.

LEMMA 4.3.1. If L contains the rule RM, then the canonical neighborhood function $f^{\mathcal{L}}$ is supplemented.

Proof. Suppose $[A] \in f^{\mathcal{L}}(\Gamma)$ and $[A] \subseteq [B]$. The inclusion means $A \to B$ belongs to every maximal L-consistent set, so $\vdash_L A \to B$. By RM, $\vdash_L \mathsf{O}A \to \mathsf{O}B$. Since $\mathsf{O}A \in \Gamma$ and Γ is maximal, $\mathsf{O}B \in \Gamma$, hence $[B] \in f^{\mathcal{L}}(\Gamma)$. ■

Remark 4.3.2. The converse direction — showing that logics *without* RM yield non-supplemented canonical models — requires care. The canonical model may accidentally satisfy frame conditions not derivable in L. See [2] for discussion. $\triangleleft$

Chapter 5

Conclusion

In this chapter we outline directions for future work arising from the results of this dissertation.

5.1 First Direction

Building on the framework developed in Chapter 2 and the results of Chapter 4, a natural next step is to...

5.2 Second Direction

The analysis in Chapter 3 suggests several open questions...

5.3 Third Direction

A longer-term goal is to...

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